

Arthur decompositions and Saito-Kurokawa existence

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Answer. No. The stated decomposition permits a Saito-Kurokawa spectral summand, but does not prove that it is nonzero. It therefore does not, by itself, construct a Saito-Kurokawa automorphic representation. This failure of implication is not a nonexistence result.

This note treats Theorem 1.1 of the supplied manuscript [M] as stated, without independently auditing its proof. The comparison with Gan [G] concerns both cuspidal Saito-Kurokawa (SK) lifts and the larger discrete packets that may contain residual representations. Page references to [M] and [G] use the supplied PDFs' printed page numbers.

1. What the decomposition actually asserts

Let $F = \mathbf{F}_q(X)$ and let G be split, connected and semisimple. For an effective level divisor D , write C_D for its finite-dimensional cuspidal space, suppressing the rational or algebraically closed coefficient field. Theorem 1.1 gives

$$C_D = \bigoplus_O C_{D,O}.$$

Here O runs over nilpotent orbits in the dual Lie algebra. A form belongs to $C_{D,O}$ when its Hecke spectrum at every place x outside D lies in the prescribed set $R_{q_x,O}$. Its normalized Satake class has the form

$$[s_x] = [\varphi_O(\text{diag}(q_x^{1/2}, q_x^{-1/2})) \cdot c_x],$$

where c_x centralizes $\varphi_O(\text{SL}_2)$ and is compact up to conjugacy at every complex embedding. The same orbit works at all good places. These are restrictions on the spectrum already present; see [M, pp. 2-3, (1.3)-(1.4)].

The missing nonvanishing statement

A direct sum may have zero summands. In fact, [M, Proposition 6.6, pp. 59-60] partitions the set Σ_D of *occurring* Hecke characters and defines

$$\begin{aligned} \Sigma_{D,O} &= \{\xi \in \Sigma_D : O_\xi = O\}, \\ e_O &= \sum_{\xi \in \Sigma_{D,O}} e_\xi, \quad C_{D,O} = e_O C_D. \end{aligned}$$

If $\Sigma_{D,O}$ is empty, the idempotent and summand are zero. Thus the theorem does not imply $C_{D,SK} \neq 0$ for any D . Allowing arbitrary finite level does not change this logical point.

2. Which orbit is Saito-Kurokawa type?

Apply the theorem first to $G = \mathrm{PGSp}_4$, the split adjoint group of this type. Its dual is Sp_4 . In the usual odd-characteristic quadratic-space realization, $\mathrm{PGSp}_4 \cong \mathrm{SO}_5$. The dual-group calculation below takes place in characteristic zero and uses the four-dimensional defining representation of Sp_4 .

Write S_d for the d -dimensional irreducible algebraic representation of the Arthur SL_2 . Gan [G, §3.8, pp. 12-13] distinguishes four orbit types:

Partition in Sp_4	Arthur SL_2 on the 4-space	Orbit/type
[1,1,1,1]	$S_1^{\oplus 4}$	Zero; tempered label
[2,1,1]	$S_2 \oplus S_1 \oplus S_1$	Minimal / long-root; SK
[2,2]	$S_2 \oplus S_2$	Short-root; different type
[4]	S_4	Regular

For the SK embedding, SL_2 acts on one symplectic plane and fixes the other. The raising operator therefore has one Jordan block of size two and two of size one. Its centralizer is

$$H_{\mathrm{SK}} = Z_{\mathrm{Sp}_4}(\varphi_{\mathrm{SK}}(\mathrm{SL}_2)) \cong \mathrm{SL}_2 \times \mu_2.$$

Consequently the local spectral set allowed by [M] has eigenvalues

$$\{\alpha_x, \alpha_x^{-1}, \delta_x q_x^{1/2}, \delta_x q_x^{-1/2}\}, \\ |\alpha_x| = 1, \quad \delta_x \in \{\pm 1\}.$$

The modulus condition holds at every complex embedding in [M]. The exponent multiset is $\{1/2, 0, 0, -1/2\}$; the representation is non-tempered at these spherical places. The local scalar δ_x here is distinct from the packet-selection sign ε_v used below.

The full SK parameter contains more information

In Gan's notation [G, §4.1, pp. 14-15], a candidate SK parameter is specified by a cuspidal representation τ of PGL_2 and a quadratic Hecke character χ :

$$\psi_{\tau, \chi} = \rho_\tau \oplus (\chi \boxtimes S_2).$$

The first summand has trivial Arthur SL_2 -action. For function-field τ , the associated unramified eigenvalues have the displayed SK shape, with $\delta_x = \chi_x(\mathrm{Frob}_x)$. Over function fields the GL_2 correspondence and purity provide the tempered rank-two input, but do not by themselves realize $\psi_{\tau, \chi}$ automorphically on PGSp_4 .

When such a lift exists, its degree-four partial L-function is $L^S(s, \pi, \mathrm{spin}) = L^S(s, \tau) L^S(s + 1/2, \chi) L^S(s - 1/2, \chi)$. Gan calls this the standard L-function in the SO_5 convention [G, p. 18]; it is the spin L-function in GSp_4 terminology, not its degree-five standard L-function.

3. Three implications that are not supplied

An allowed orbit need not occur

The theorem has the form: every existing cuspidal Hecke system has a unique allowed orbit label. Existence would require the converse assertion that the particular label $[2,1,1]$ actually occurs. The construction of the projectors in §1 supplies no positive dimension estimate and no vector on which the SK projector acts nontrivially. It is logically compatible with the theorem that the SK summand vanishes at every level.

A Galois parameter is not an automorphic realization theorem

[M, Proposition 6.5, p. 58] does use Vincent Lafforgue's global parameters. It first fixes an already occurring Hecke character and uses its nonzero eigenspace to obtain an excursion character. The resulting direction is

existing automorphic eigensystem $\rightarrow$ global Galois parameter.

This does not assert that every proposed parameter $\psi_{\tau,\chi}$, or its associated Galois parameter, has a nonzero automorphic eigenspace. This distinction persists even after the GL_2 input τ is known to exist. See also [V] for the scope of the global parametrization.

An orbit label is coarser than an Arthur packet

The summand $C_{D,[2,1,1]}$ is indexed only by the Arthur SL_2 -type. The displayed decomposition does not index it by pairs (τ,χ) , construct local packets, impose a global multiplicity formula, or identify it with the space of Siegel-CAP lifts. Such an identification needs additional theorems. In [GLR], the orbit decomposition in Conjectures 3.4.5-3.4.6 is distinguished from the stronger parameter-support assertions in §3.6; neither a support inclusion nor a decomposition says that every candidate contributes.

How this answers the question for GSp_4

GSp_4 is reductive with a positive-dimensional center, so it is not directly among the semisimple groups of [M, Theorem 1.1]. PGSp_4 is in scope. A cuspidal representation on PGSp_4 pulls back along $\mathrm{GSp}_4 \rightarrow \mathrm{PGSp}_4$ to a cuspidal representation with trivial scalar central character; cuspidality is preserved because the unipotent radicals are unchanged. Square-integrability here is understood modulo the center.

Twisting this pullback by $\mu \circ \mathrm{sim}$, for a unitary Hecke character μ , gives central character μ^2 , since $\mathrm{sim}(zI) = z^2$. Thus a PGSp_4 existence theorem would already give examples on GSp_4 . It would not, without more input, settle arbitrary prescribed central characters. This group-scope issue is separate from the fundamental missing nonvanishing statement.

Precise consequence. If a cuspidal SK lift on PGSp_4 over F is independently constructed, the decomposition locates its finite-level vectors in the $[2,1,1]$ summand. The decomposition alone does not perform that construction.

4. What Gan adds over number fields

Gan explicitly fixes a **number field** on p. 1 of [G]. The following is therefore a comparison theorem in that setting, not an unqualified theorem over the user's function field.

Local packets and the global sign

For a pair (τ, χ) , let T_τ be the finite set of places where τ_v is discrete series. Gan constructs the local packet $\{\pi_v^+\}$ if $v \notin T_\tau$, and $\{\pi_v^+, \pi_v^-\}$ otherwise. At almost all places π_v^+ is the spherical SK quotient. An abstract global packet member is

$$\pi^\varepsilon = \otimes'_v \pi_v^{\varepsilon_v}, \quad \varepsilon_v = +1 \text{ for } v \notin T_\tau.$$

In Gan's constructed subspace $A(\tau, \chi)$, the multiplicity formula is [G, §4.3 and Proposition 5.8, pp. 15-18]:

$$\begin{aligned} m_A(\pi^\varepsilon) &= 1 \iff \prod_v \varepsilon_v = \varepsilon(1/2, \tau \otimes \chi); \\ m_A(\pi^\varepsilon) &= 0 \text{ otherwise.} \end{aligned}$$

This formula comes with an actual global construction: a suitable Waldspurger representation of the metaplectic double cover of SL_2 is lifted to split SO_5 . Gan proves nonvanishing and automorphy [G, §5.7-5.8]; Theorem 5.9 identifies the cuspidal parts of these spaces with the full Siegel-CAP space. An orbit decomposition supplies none of these steps.

Discrete occurrence and cuspidality are different

Among the realized members, a lift is cuspidal if some $\varepsilon_v = -1$ or $L(1/2, \tau \otimes \chi) = 0$. The exceptional all-plus member with nonzero central L-value is residual. Gan gives the cuspidality statement in Proposition 5.8; the noncuspidal exception is explicit in Schmidt [S, introduction and Theorem 3.1].

Packet situation	What follows in this number-field recipe
T_τ empty; root number -1	Only the all-plus abstract member exists; its multiplicity is zero.
T_τ empty; central L-value $\neq 0$	Root number is $+1$; the realized singleton is residual, so its cuspidal contribution is zero.
A minus sign is available and chosen; global product is correct	The member occurs with multiplicity one and is cuspidal.

The first row is an obstruction mechanism for input satisfying those hypotheses, not an assertion that such input occurs for every field. Gan §1.9 (p. 4) discusses the corresponding empty automorphic Waldspurger contribution.

The central L-value condition should not be reversed: its nonvanishing does not force the SO_5 lift to be cuspidal. Gan's discussion in §2.16-2.17 (p. 9) explains that the lift to dimension three has a central-value obstruction, whereas the relevant dimension-five nonvanishing is governed by local theta lifts. Cuspidality also involves first occurrence in the theta tower.

5. What a function-field existence proof needs

The negative answer concerns the implication from [M], not the possibility of SK representations in positive characteristic. Function-field theta theory does exist: for example, Xiong [X] proves a Siegel-Weil formula and convergence results for classical groups in odd characteristic. That result is not itself a theorem realizing all SK packets.

A route modeled on Gan must establish, for the chosen function field and characteristic: (i) suitable cuspidal PGL_2 input and a nonzero automorphic metaplectic representation with the required local signs; (ii) nonvanishing and the correct local behavior of its theta lift to the split rank-two target; and (iii) cuspidality, or else identification of a residual contribution. If one also wants a complete classification, one must establish the packet multiplicities and exhaustion. None follows just from [M, Theorem 1.1].

A useful conditional existence criterion

Suppose, as additional hypotheses, that Gan's local/global construction and cuspidality criterion hold over the chosen function field, and that a cuspidal τ of PGL_2 has at least two discrete-series places. For any quadratic χ , a nonempty subset $E \subset T_\tau$ can be chosen so that

$$(-1)^{|E|} = \varepsilon(1/2, \tau \otimes \chi).$$

For root number -1 take one place; for root number $+1$ take two. Choose the minus member exactly on E . The assumed multiplicity formula gives automorphic occurrence, and the minus sign gives cuspidality. Pullback then gives a GSp_4 representation. This is a conditional deduction illustrating the missing input; the required function-field lifting results and τ are assumptions here.

No conflict with the generic Ramanujan corollary

[M, Corollary 1.2, pp. 4 and 60-61] applies to split adjoint absolutely simple groups and requires a generic unramified component. It concludes that all unramified components are tempered. SK spectral type has eigenvalue moduli $q_x^{\pm 1/2}$, so a hypothetical cuspidal representation in this summand cannot have a generic unramified component. This excludes it from the corollary's hypothesis, not from the cuspidal spectrum. Global genericity would imply local genericity at every place.

Historical comparison. The early Ichino-Ikeda draft [II, §14, pp. 71-72] makes its unramified SK packet discussion conditional on a function-field multiplicity formula (genus one, odd characteristic). This illustrates the distinction; it establishes neither nonexistence nor that all function-field cases are open.

Conclusion. The additional assertion needed for SK spectral existence is $C_{D,[2,1,1]} \neq 0$ for some D . Identifying the resulting representations as SK lifts in the stronger Siegel-CAP/theta sense requires further input. The supplied decomposition establishes the allowed spectral form and its consistency across places; it establishes neither assertion of existence.

References and verification guide

Primary sources were checked on 8 October 2026. The two supplied PDFs are the principal sources; external references clarify the distinction between parametrization, lifting, and positive-characteristic theta theory.

[M] OpenAI. *Ramanujan-Arthur Decompositions of Cuspidal Functions at Full Finite Level.*

Supplied manuscript, title-page date 24 September 2026, 64 pp. File: paper.pdf. Key locators: Theorem 1.1, pp. 2-3; Corollary 1.2, p. 4; Proposition 6.5, pp. 58-59; Proposition 6.6, pp. 59-60; §6.5, pp. 60-61.

[G] Wee Teck Gan. *The Saito-Kurokawa space of $PGSp_4$ and its transfer to inner forms.* Supplied 32-page notes, gan.pdf. Key locators: number-field convention, p. 1; §1.9, p. 4; §3.8, pp. 12-13; §4.1-4.3, pp. 14-16; Proposition 5.8 and Theorem 5.9, p. 18. [Online copy of the notes.](#)

[GLR] Dennis Gaitsgory, Vincent Lafforgue and Sam Raskin. *Tempered vs generic automorphic functions and the canonical filtration on automorphic functions.* arXiv:2603.26925v2. Compare Conjectures 3.4.5-3.4.6 and §3.6 on Arthur parameters. [Versioned text.](#)

[V] Vincent Lafforgue. *Introduction to chtoucas for reductive groups and to the global Langlands parameterization.* arXiv:1404.6416. Used for the direction and scope of the parameter attachment; the relevant use in the supplied manuscript is [M, Proposition 6.5]. [Author's introductory article.](#)

[S] Ralf Schmidt. *The Saito-Kurokawa Lifting and Functoriality.* Introduction, p. 1, and Theorem 3.1 with proof, pp. 10-11. Explicitly distinguishes cuspidal lifts from the all-plus, nonzero-central-value exception; the global field is a number field. [Author PDF.](#)

[X] Wei Xiong. *On the Siegel-Weil formula for classical groups over function fields.* Journal of Number Theory 215 (2020), 52-97; arXiv:1806.02049v5. See the abstract and introduction for odd-characteristic scope. [Preprint](#); [published article.](#)

[II] Atsushi Ichino and Tamotsu Ikeda. *On the periods of automorphic forms on special orthogonal groups and the Gross-Prasad conjecture.* Early 83-page draft, Part IV: §12, p. 54, and §14, pp. 71-72. Cited only for its explicitly conditional function-field discussion, not as a present-day nonexistence or open-problem claim. [Early author draft.](#)

Reading order. For the decisive logical point, read [M, p. 59]. For the correct orbit, read [G, p. 13]. For the extra existence and cuspidality input, compare [G, pp. 15 and 18].